PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T37 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

8 classified items • 11 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T37 • Expertise Q16+ • 8 items • 11 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Transformations of Graphs

Unit 2 • Expertise • 11 marks • 8 item(s)

June 2024 | 4MA1-2H Q25 | 2 marks

Horizontal translation

November 2024 | 4MA1-2H Q19 | 1 marks

Vertical scale or x-axis reflection

November 2024 | 4MA1-2H Q19 | 1 marks

Vertical translation

June 2022 | 4MA1-2H Q23 | 1 marks

Horizontal scale or y-axis reflection

November 2021 | 4MA1-2H Q21 | 1 marks

Transform an explicit polynomial and simplify

November 2021 | 4MA1-2H Q21 | 1 marks

Translation vector applied to $y=f(x)$

November 2021 | 4MA1-2H Q21 | 1 marks

Describe from two curve equations

November 2021 | 4MA1-2H Q21 | 3 marks

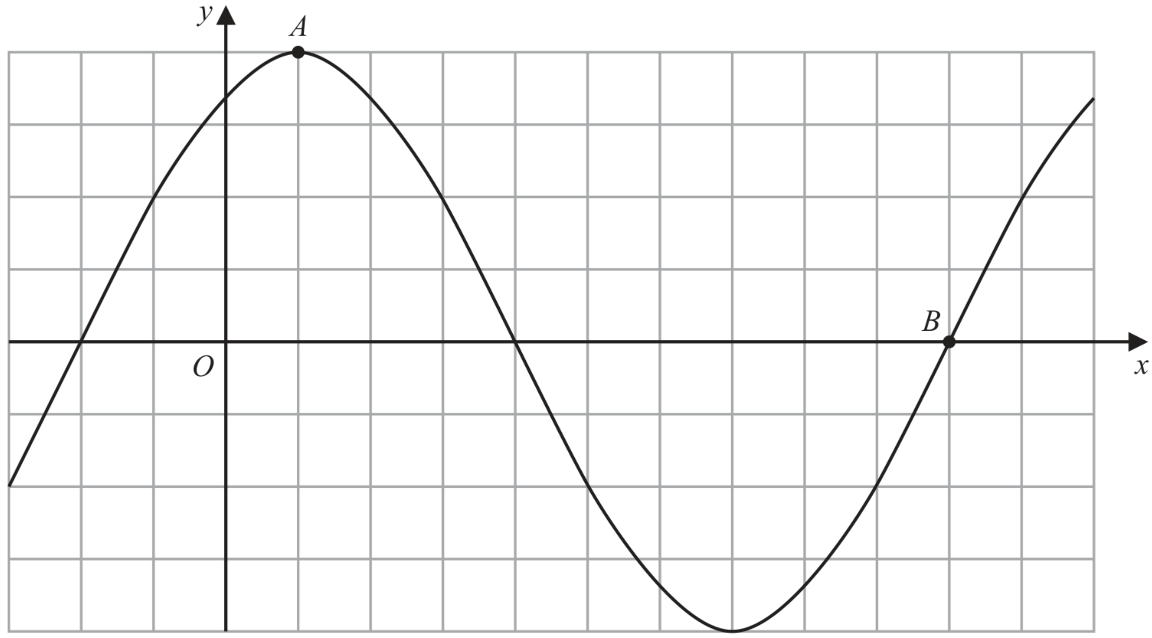
Combined feature mapping

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q25 **2 marks**

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)°$



(i) Find the coordinates of the point A

(.......................... , ...)

(1)

(ii) Find the coordinates of the point B

(.......................... , ...)

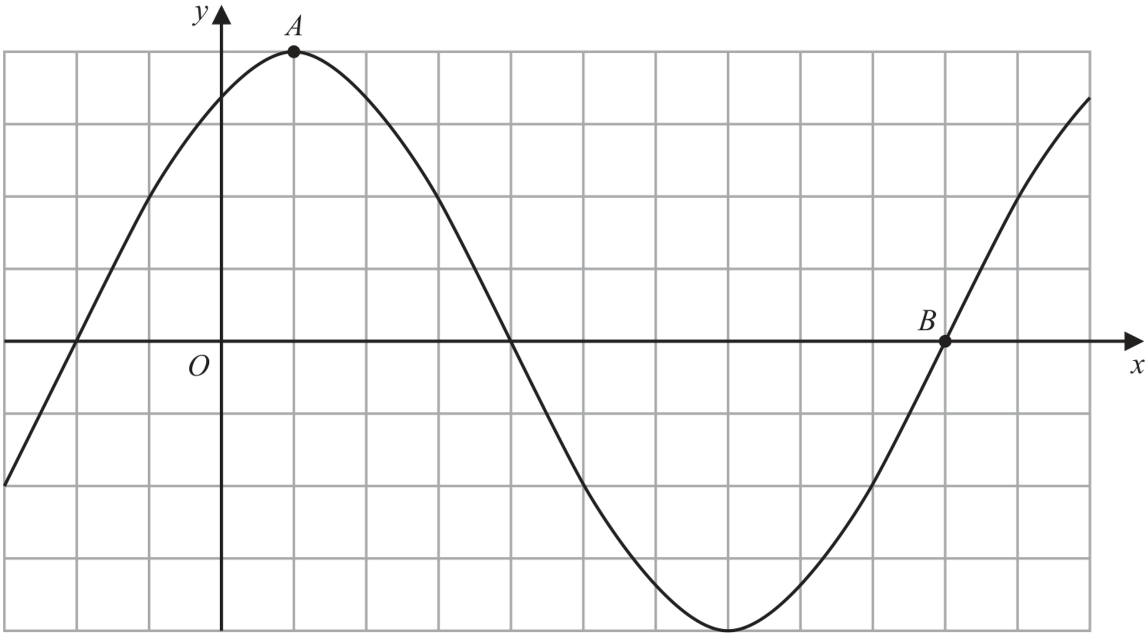
(1)

Continue your method **2 marks**

WORKING SPACE

4MA1-2H Q25 **2 marks**

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)°$



(i) Find the coordinates of the point A

(.......................... , ...)
(1)

(ii) Find the coordinates of the point B

(.......................... , ...)
(1)

Transformed sine graph **2 marks**

Maximum

$$x + 60 = 90 \Rightarrow A = (30, 2).$$

Axis crossing

$$x + 60 = 360 \Rightarrow B = (300, 0).$$

Final answer

$$A = (30, 2), \quad B = (300, 0)$$

4MA1-2H Q19 M02 **1 marks**

19 A curve has equation $y = \mathrm{f}(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (5, 4)

(ii) $y = 3\mathrm{f}(x)$

(.............................. , ...)
(1)

WORKING SPACE

Stretch graph values **1 marks**

M02 – Stretch graph values

Multiply the output
For $y=3f(x)$, the x-coordinate stays 5 and the y-coordinate becomes $3(4)=12$.

Final answer
$(5,12)$

4MA1-2H Q19 M03 **1 marks**

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (5, 4)

(iii) $y = f(x) - 7$

(.............................. , ...)
(1)

WORKING SPACE

Shift a graph vertically **1 marks**

M03 – Shift a graph vertically

Subtract 7 from y

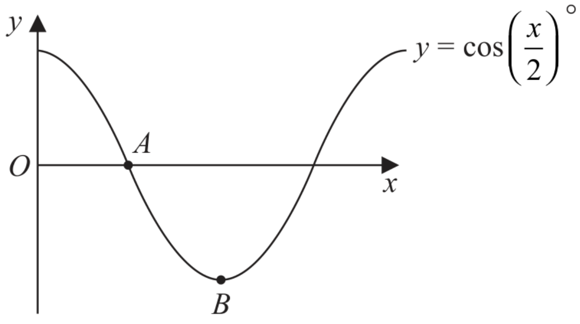
For y=f(x)-7, (5,4) becomes (5,4-7)=(5,-3).

Final answer

(5,-3)

4MA1-2H Q23 Q23(ii) **1 marks**

23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



(ii) Find the coordinates of the point B

(.............................. , ...)

(1)

Continue your method **1 marks**

WORKING SPACE

4MA1-2H Q23 Q23(ii) **1 marks**

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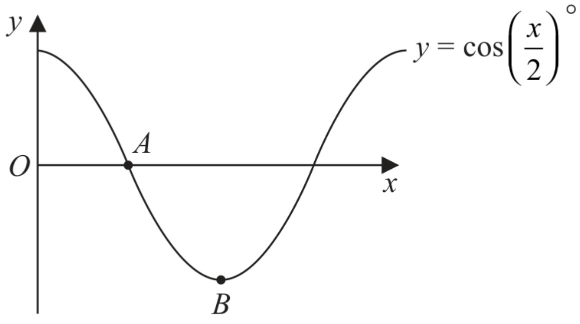


(ii) Find the coordinates of the point B

(.............................. , ...)

(1)

Chapter: Transformations of Graphs **1 marks**

Q23(ii) – Chapter: Transformations of Graphs

Family: Map a known graph feature • Variant: Horizontal scale or y-axis reflection

Method

At the first minimum, x/2 = 180 degrees, so x = 360 degrees and y = -1; hence B = (360, -1).

Why: Use the first minimum of the cosine graph in degree mode.

Final answer

B: B = (360, -1)

21 The curve **C** has equation $y = \text{f}(x)$ where $\text{f}(x) = 9 - 3(x + 2)^2$
The point A is the maximum point on **C**.

(a) Write down the coordinates of A.

(.......................... , ...)

(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

....

(1)

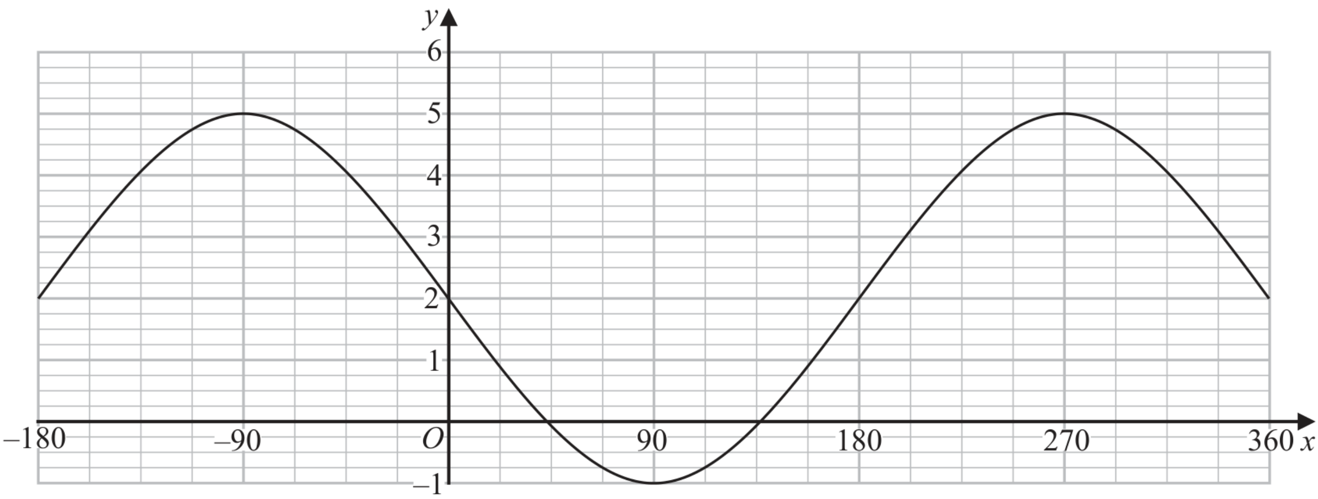
The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

....

(1)

The graph of $y = a\cos(x - b)° + c$ for $-180 \leqslant x \leqslant 360$ is drawn on the grid below.



(d) Find the value of a, the value of b and the value of c.

a = ...

b = ...

c = ...

(3)

Worked solution

November 2021 | Question 21 | 1 marks

Family: Write the transformed equation • Variant: Transform an explicit polynomial and simplify

Method / working
1. f(x)=9-3(x+2)^2
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. This is in completed square form, so the maximum point is
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. (-2,9)
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. For the translation
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. \begin{pmatrix}4\\0\end{pmatrix}
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
6. the graph moves 4 units right. Replace \(x\) by \(x-4\):
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
7. y=9-3((x-4)+2)^2
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
8. y=9-3(x-2)^2
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9. The curve \(T\) has equation
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10. y=3(x+2)^2-9
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
11. This is
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
12. y=-\left(9-3(x+2)^2\right)
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
13. So \(C\) is reflected in the \(x\)-axis.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
14. For the cosine graph, the maximum is 5 and the minimum is \(-1\).
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
15. a=frac{5-(-1)}{2}=3
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
16. c=frac{5+(-1)}{2}=2
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
17. For
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
18. y=a\cos(x-b)^\circ+c
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
19. a maximum occurs when \(x-b=0\). The graph has a maximum at \(x=270^\circ\), so
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
20. b=270
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
21. Answer: \(A=(-2,9)\), \(y=9-3(x-2)^2\), reflection in the \(x\)-axis, \(a=3,\ b=270,\ c=2\)
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
Final answer: A=(-2,9), y=9-3(x-2)^2, reflection in the x-axis, a=3,\ b=270,\ c=2

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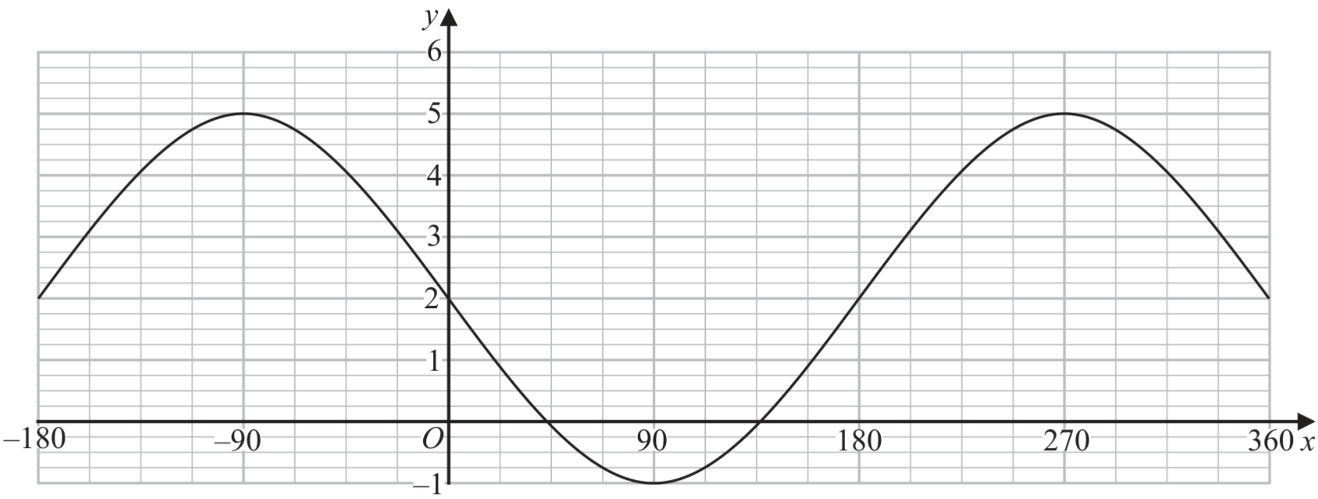

(d) Find the value of a, the value of b and the value of c.

a = ...

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c = ...

(3)

Worked solution

November 2021 | Question 21 | 1 marks

Family: Write the transformed equation • Variant: Translation vector applied to y=f(x)

Method / working
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Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
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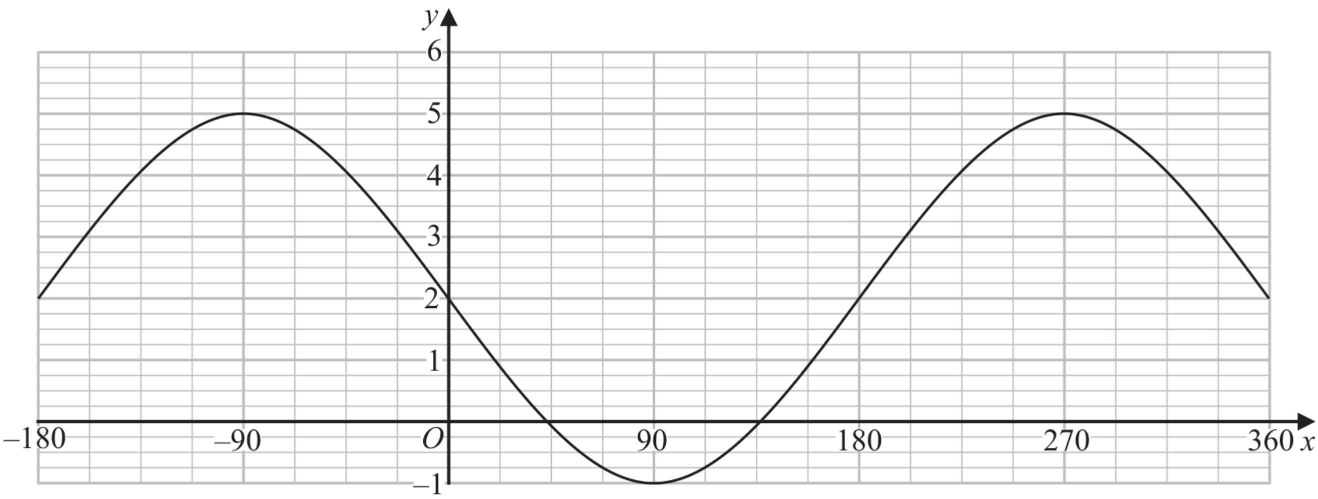

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Worked solution

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Family: Infer or describe the graph transformation • Variant: Describe from two curve equations

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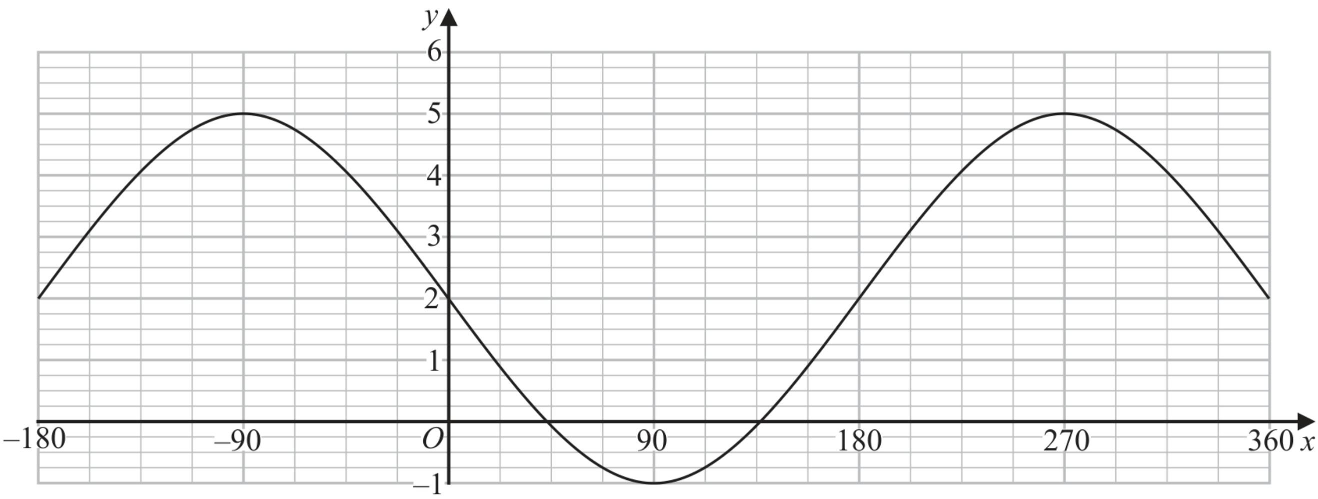

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Worked solution

November 2021 | Question 21 | 3 marks

Family: Map a known graph feature • Variant: Combined feature mapping

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